



Service Tool Instruction

Service Tool Instruction
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Description

Injector Tester

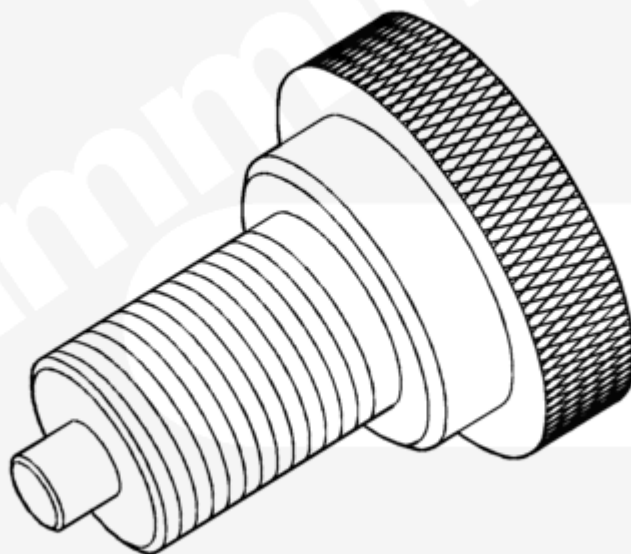
Purpose

This service tool instruction provides information for using the QSK19 injector tester, Part Number 3825165, and the QSK45 and QSK60 injector tester, Part Number 3163194. The purpose of the injector tester is to detect defective or malfunctioning injector(s) in the QSK19, QSK45 and QSK60 engines.

The injector tester screws into a 27 mm threaded hole in the top of the production cast iron valve cover. The injector tester is gradually rotated, after the engine has been operating at normal coolant temperature 90 °C [180 °F], **clockwise** by hand. This action pushes the injector rocker lever and the injector plunger down far enough to shut off the fuel supply to the injector. Shutting off the fuel supply to a good working injector will result in a drop in the engine speed and an increase in fuel rail pressure. A defective or malfunctioning injector will show no change in engine speed or rail pressure.

⚠ CAUTION ⚠

Use this injector tester only with a production cast iron valve cover. Do not use with a plastic or aluminum valve cover. This will result in damage to the valve cover(s).



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Table 1, QSK19 Injector Tester, Part Number 3825165

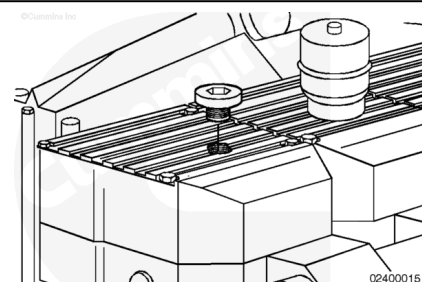
Item Number	Part Number	Description	Quantity
(Shown above)	3825165	QSK19 injector tester	1

Table 2, QSK45 and QSK60 Injector Tester, Part Number 3163194

Item Number	Part Number	Description	Quantity
(Shown above)	3163194	QSK45 and QSK60 injector tester	1

After the engine is warmed up to the normal coolant temperature of 90 °C [180 °F], shut off the engine.

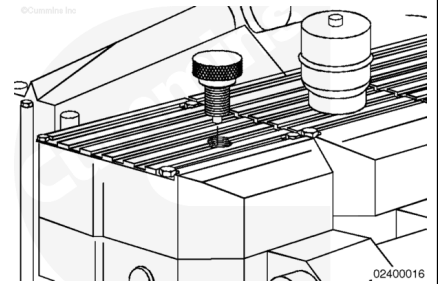
Use a 12 mm hexagon socket or Allen wrench to remove the existing plug(s) from the top of the cast iron valve cover(s).



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⚠ CAUTION ⚠

Do not use any tools when tightening the injector tester(s) to the valve cover(s). Failure to do so can cause damage to the injector.

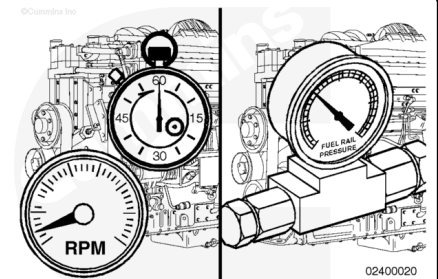


Note : One injector tester, Part Number 3825165 is required for the QSK19 series engine. Seven injector testers, Part Number 3163194 are required for the QSK45 series engine and nine are required for the QSK60 series engine.

Gradually rotate the injector tester **clockwise** by hand into the threaded hole in the valve cover until the injector plunger is pushed down far enough to shut off the fuel supply to the injector.

⚠ CAUTION ⚠

The injector tester is not intended for use above engine idle speed. Do not use above 900 rpm engine speed. Failure to do so will result in damage to the engine.



Refer to the Troubleshooting and Repair Manual, QSK19 Base Series Engine, Bulletin 3666098, for the injector test procedures.

Refer to the Troubleshooting and Repair Manual, QSK45 and QSK60 Base Series Engines, Bulletin 3666261, for the injector test procedures.

After injector testing, remove the injector tester(s) from the valve cover(s) and install the plug(s).

Tighten the plug.

Torque Value: 45 n·m [35 ft-lb]

